1. The primary valences are non-directional while the secondary valences are directional. The geometry of the complex is determined by the spacial arrangement of the groups which satisfy the secondary valence. For example, if a metal ion has a secondary valence of six, it has an octahedral geometry. If the secondary valence is 4, it has either tetrahedral or square planar geometry.

The following table illustrates the Werner's postulates.

Complex	Groups satisfy the secondary valence (non-ionaisable, inner coordination sphere)	No. of ionisable Cl^- ions in the complex (outer coordination sphere)	No. of moles of AgCl formed = no. of moles of ionisable Cl^-
$CoCl_3.6NH_3$	6 NH_3	3 Cl^-	3 AgCl
$CoCl_3.5NH_3$	5 NH_3 & 1 Cl^-	2 Cl^-	2 AgCl
$CoCl_3.4NH_3$	4 NH_3 & 2 Cl^-	1 Cl^-	1 AgCl
$CoCl_3.4NH_3$	4 NH_3 & 2 Cl^-	1 Cl^-	1 AgCl

5.2.1 Limitations of Werner's theory:

Even though, Werner's theory was able to explain a number of properties of coordination compounds, it does not explain their colour and the magnetic properties.

Evaluate yourself 1:

When a coordination compound $CrCl_3.4H_2O$ is mixed with silver nitrate solution, one mole of silver chloride is precipitated per mole of the compound. There are no free solvent molecules in that compound. Assign the secondary valence to the metal and write the structural formula of the compound.

5.3 Definition of important terms pertaining to co-ordination compounds

5.3.1 Coordination entity:

Coordination entity is an ion or a neutral molecule, composed of a central atom, usually a metal and the array of other atoms or groups of atoms (ligands) that are attached to it. In the formula, the coordination entity is enclosed in square brackets. For example, in potassium ferrocyanide, $K_4[Fe(CN)_6]$, the coordination entity is $[Fe(CN)_6]^{4-}$. In nickel tetracarbonyl, the coordination entity is $[Ni(CO)_4]$.

5.3.2 Central atom/ion:

The central atom/ion is the one that occupies the central position in a coordination entity and binds other atoms or groups of atoms (ligands) to itself, through a coordinate

XII U5 Coordination jagan.indd 133 04-01-2021 13:24:11